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# [Forest Sampling & Sampling Techniques]

## 1. Census v Sampling

Census measures each and every unit in a population while Sampling measures a small proportion of the units in a population.

## 2. Objective of Sampling

"to Secure a Sample which will represent the population and reproduce the important Characteristics of the population under Study as Closely as possible"

## 3. Population

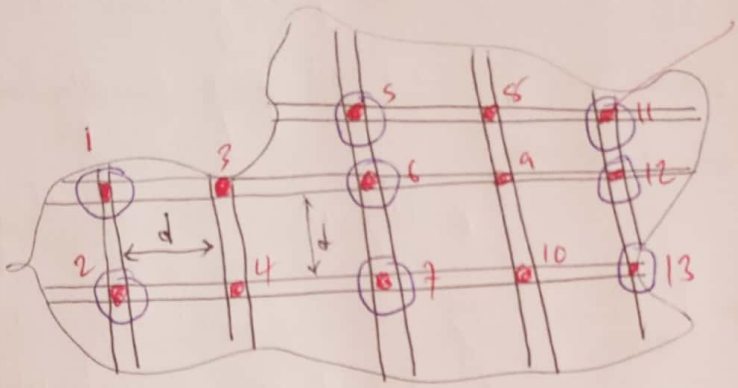
The word population is the aggregates of units from which a Sample is Chosen.

The aggregate of trees in the forest from which we are going to Chose a Sample [for a forest]

## 4. Sampling Units

Sampling units may be administrative Units or natural Units like topographic Sections and Subcompartments or may be artificial units like Strips of Certain width, or plots of a definite Shape and Size. The Unit must be a well defined element or group of elements identifiable in the forest area on which observations

On the Characteristics under Study could be made. The population is thus subdivided into suitable Units for the purpose of Sampling and these are Called Sampling Units.




artificial unit

14 Sampling Units

## 5. Sampling Frame

A list of Sampling Units will be Called a 'frame'

## 6. Sample

One or more Sampling Units Selected from a population according to some Specified procedure will Constitute a Sample.

Sampling from: 1 (2) 3 (4) 5 (6) 7 (8) 9 (10) 11 (12) 13

Procedure: Every even Value

N.B: The procedure Can Vary eg randomly, etc.

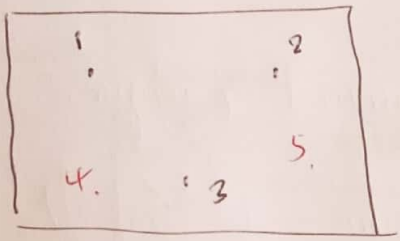
# 7. Sampling Intensity

Intensity of Sampling is defined as the ratio of the number of units in the Sample to the number of units in the population.

$$\text{Sampling Intensity} = \frac{\#\text{ of units in Sample}}{\#\text{ of units in population}}$$

$n = 10{,}000$
$S = 100$

$$= \frac{100}{10000} = 0.01 \text{ or } 1\%$$




Population = P
# of Sample = n
# of units in each Sample = m

# of units in a Sample = m x n

$$SI = \frac{m \times n}{P} \times 100\%$$

SI can be increased by increasing m or n or by reducing P. However, P is Constant

$n = 3$
$m = 10$

# of units in Sample = $3 \times 10 = 30$ $\quad SI = \frac{30}{10000} = 0.003$

$n = 5$
$m = 10$

$= 5 \times 10 = 50$ $\quad SI = \frac{50}{1000} = 0.005$

1] $n = 3$

from 10 → $n = 100$

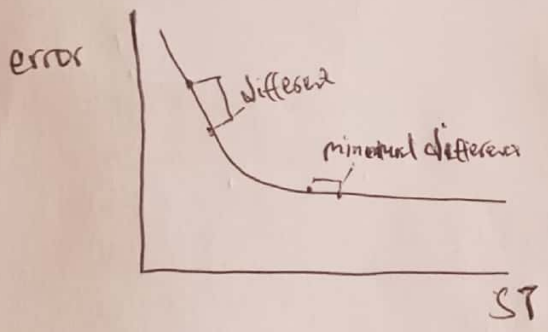
# of units in Sample $n \times n = 3 \times 100 = 300$

$$= \frac{300}{10,000} = \boxed{0.03}$$

How does SI Matter to us

NB: If SI is 100% then we are measuring each and every unit in a population, which is equivalent to a Census.

Error and SI




As the SI increase the error reduces. Take SI at a point which error decrease Considerably and the amount of benefit is not decreasing.

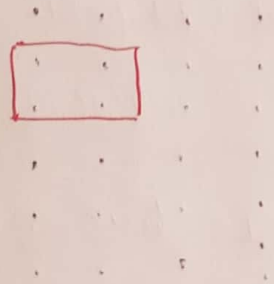
8. Kinds of Plots

- Circula
- Rectangular
- Strips
- Topographical Units: Used in hills

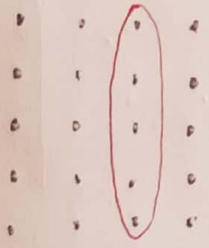
* When do we take each of these units above?



Circular plot is prefered because our samples do not have any directionality



Here you prefer to take a rectangular plot, because the individual units are arranged in a rectangular form

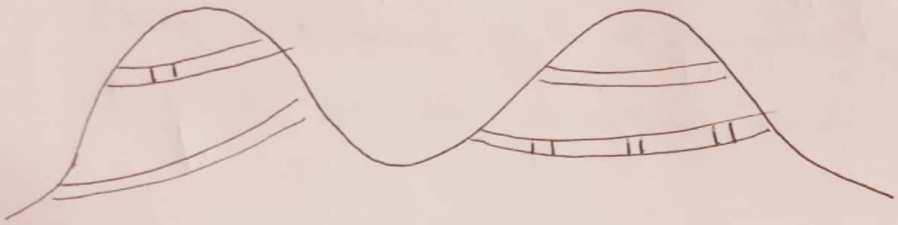


Strips can also be prefered (Survival rate)

1
2
3
4 — Randomly pick 4 and take every 10th row
5
6
7
8 4th, 14, 24, 34 ...
9
10 SI = 10%

This is better than rectangular plots because you have to take Coordinates but for strips you go directly to the field and measure the parameters

Topographical plot: for hills



## 9. Kinds of Sampling

1. Simple random Sampling
2. Systematic Sampling
3. Stratified Sampling
4. Multi Stage Sampling
5. [PPS] Probability proportional to Size Sampling

### Simple random Sampling

A Sampling procedure such that each possible Combination of Sampling Units out of the population has the same Chance of being Selected is referred to as Simple random Sampling.

Selection: lottery, random numbers

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Pick randomly
Hence list all the individuals.

## Systematic Sampling

Systematic Sampling employs a simple rule of selecting every Kth Unit with a number Chosen at random from 1 to K as the random start.

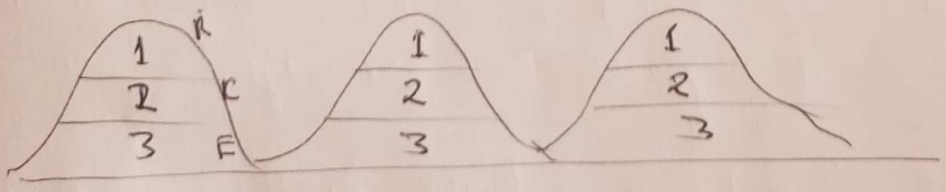
## Stratified Sampling

"The basic idea in Stratified random Sampling is to divide a heterogeneous population into sub-populations usually Known as Strata, each of which is internally homogeneous in which case a precise estimate of any Stratum mean Can be obtained based on a Small Sample from that Stratum and by Combining Such estimates, a precise estimate of the who population Can be obtained."

* In Case of forest we can take values of

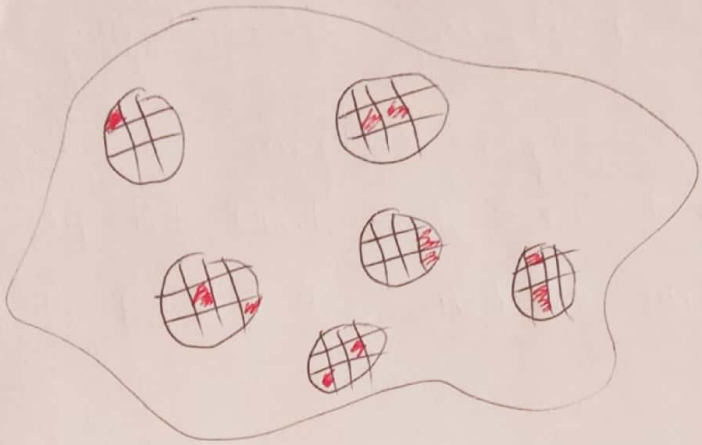
trees Close to the river
trees towards the edges of the forest
trees at the centre of the forest

} mean height of these Strata




## Multistage Sampling

The procedure of first selecting large Sized Units and then Choosing a Specified number of Sub-Units from the Selected Units Known as Sub-Sampling



# PPS Sampling

When units vary in size and the variable under study is directly related with the size of the unit, the probabilities may be assigned proportional to the size of the unit. This type of sampling where the probability of selection is proportional to the size of the unit is known as PPS sampling. eg point sampling - the probability of selecting a larger size tree is greater than probability of selecting a smaller sized tree.